

# Raymond He

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## EDUCATION

### Carnegie Mellon University — B.S. in Computer Science

Expected May 2028

GPA: 3.9/4.0 | Dean's List with High Honors | USA Computing Olympiad (USACO) Gold DivisionPittsburgh, PA

- **Coursework:** Data Structures & Algorithms, Computer Systems, Theoretical CS; Fall 2026: Deep Learning, Natural Language Processing, Parallel Computer Architecture & Programming

## EXPERIENCE

### Roswell Studios — AI Engineering Intern

Jun 2026 – Present

AI accessibility remediation platform for Shopify storefronts

New York, NY

- Shipped an end-to-end platform that detects WCAG 2.1 AA violations (Playwright + axe-core), generates source-code fixes with Claude-based agents, and automatically opens pull requests against client repositories, supporting hundreds of eCommerce brands.
- Architected a *multi-agent orchestration pipeline* — a central orchestrator coordinating planner, implementer, and verifier agents via structured tool calling — tracing violations to Shopify Liquid, CSS, and JavaScript sources and applying minimal root-cause fixes with approximately 98% validation accuracy, backed by an *automated regression eval suite*.
- Built a closed-loop validation system with DOM-stability detection to produce consistent scans on dynamic storefronts, rescan production pages after deployment, and classify each fix as confirmed, failed, or regressed.
- Developed multi-tenant infrastructure: resumable breadth-first crawling, deduplicated SQLite storage, GitHub Actions CI/CD, webhook-based deployment monitoring, and a real-time admin dashboard.

### Sequoia Management LP — Quantitative Research Intern

Jun 2026 – Present

- Discovered a short-side hedge for intraday SPY strategies that raised Sharpe ratio from 0.95 to 1.19, cut max drawdown from 19% to 14.5%, and achieved an 80% hit rate on the 10 worst SPY trading days.
- Developed intraday momentum and breakout signals from minute-bar data using VWAP and volatility features; built a Python backtesting engine with volatility-targeted sizing, commission modeling, and automated P&L/AUM checks guarding against look-ahead bias.
- Extending research to a multi-asset futures trend-following model with validated continuous-contract and roll-adjustment data pipelines.

## RESEARCH & PROJECTS

### FastGraph — Researcher & Co-Author

Jun 2024 – Jan 2026

- Co-authored *FastGraph*, a manuscript submitted to Elsevier's *Array*, on differentiable k-nearest-neighbor algorithms for graph neural network workloads, achieving 20–50× speedups over FAISS and ScaNN.
- Implemented custom CUDA kernels with PyTorch-compatible gradients; benchmarked on datasets exceeding 10M points (NVIDIA RTX 4080).

### LLM-Powered Maritime Freight Matcher — Argo Oriental Shipping Ltd.

Jun 2026

- Engineered a three-stage Anthropic API pipeline extracting structured cargo and vessel data from 10,000+ maritime emails at ~\$0.005/email, reaching ~99% extraction accuracy via *the Anthropic Batch API*, *structured outputs with JSON-schema validation*, and deterministic post-processing.
- Deployed a matching engine generating ~200 cargo-to-vessel matches per day with automated scheduling, SQLite persistence, and a searchable web dashboard used by *dozens of employees*.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Java, SQL, JavaScript, SML

**ML & Data:** PyTorch, TensorFlow, scikit-learn, CUDA, NumPy, Pandas, *multi-agent orchestration*, *LLM evals*, Anthropic/OpenAI APIs

**Systems & Tools:** Git, GitHub Actions, AWS, GCP, SQLite, Node.js, Express, Playwright